

KUNAL M SHENDE

Software Engineer | IoT Developer | Cybersecurity Researcher

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SUMMARY

Computer Science undergraduate with 2+ years of hands-on experience in embedded systems, IoT hardware security, and full-stack web development. Built ZeroTrace, an ESP32-S3 cybersecurity audit device featuring real-time RF spectrum analysis and USB-HID automation, which won 2nd Prize among 50+ teams at a national-level competition. Holds an Indian Design Patent filing (Application No: 499678-001). Skilled in vulnerability assessment, penetration testing, and delivering scalable web platforms with Python Flask, HTML5/CSS3, and cloud integrations.

TECHNICAL SKILLS

Languages: C, Embedded C, C++, Python, TypeScript, Bash

IoT and Hardware: ESP32, ESP32-S3, Arduino, Raspberry Pi, FreeRTOS, SPI, I2C, UART

Frontend: HTML5, CSS3, Bootstrap

Backend: Python Flask, WebSockets, Firebase

Databases: MySQL, Supabase, Firebase Firestore

Security Tools: Burp Suite, Nmap, Wireshark, Metasploit, OWASP Top 10

DevOps and Tools: Git, GitHub, Docker, Vercel, Google Cloud, VS Code, Postman

EXPERIENCE

Independent Security Researcher | *Freelance* | January 2024 - Present

- Architected 3 secure IoT prototypes on ESP32-S3 and ATmega328P, reducing firmware attack surfaces by 40% through memory protection techniques.
- Identified 15+ critical vulnerabilities across 10+ controlled environments using OWASP and NIST assessment methodologies.
- Lowered network security anomalies by 25% through traffic analysis and proactive system hardening across 5 IoT deployments.

Web Exploit Hunting Intern | *EduSkills and AICTE* | June 2026 - August 2026

- Uncovered 12 security flaws including SQL injection and XSS across 8 web applications during structured penetration testing exercises.
- Accelerated patch deployment timelines by 30% by compiling detailed remediation reports for the development team.

Cybersecurity Intern | *Palo Alto Networks / EduSkills* | January 2026 - March 2026

- Examined 500+ packet captures and firewall configurations, strengthening endpoint protection across 3 multi-cloud environments.
- Earned 4 industry certifications (SecOps, Network Security, Cybersecurity, Cloud Security) during the internship period.

Java Full Stack Developer Virtual Intern | *EduSkills and AICTE* | October 2025 - December 2025

- Completed hands-on coursework in full-stack application development, database integration, and backend service fundamentals.

Cloud Virtual Intern | *AWS Academy* | April 2025 - June 2025

- Studied cloud computing architectures, covering fundamental AWS infrastructure services including EC2, S3, VPC, and IAM.

Android Developer Virtual Intern | *Google for Developers* | January 2025 - March 2025

- Explored native Android application fundamentals, Material Design interface principles, and local SQLite data storage.

PROJECTS

ZeroTrace - Cybersecurity Audit Device | *ESP32-S3, FreeRTOS, C++, USB-HID* | [\[GitHub Repo\]](#) | 2026

- Created an all-in-one embedded audit companion with 4 security modules: RF scanning, Wi-Fi auditing, RSSI foxhunting, and USB-HID injection.
- Achieved 60 FPS display rendering by pinning FreeRTOS tasks across dual CPU cores, cutting frame latency by 35%.

- Won 2nd Prize at TECHTANTRA 2K26 national competition among 50+ teams; filed Design Patent (No: 499678-001).

Sentinel - HID Attack Detection Framework | *Python, Windows API, WMI* | 2026

- Launched a real-time USB threat detection system that neutralizes rogue HID devices with sub-10ms response time.
- Captured 100% of simulated BadUSB attack vectors through automated forensic snapshotting in controlled testing.

MicroFlasher - Web Serial Flashing Platform | *TypeScript, Next.js, Web Serial API* | 2026

- Released a zero-install browser-based firmware flasher, eliminating native toolchain setup and cutting developer onboarding time by 70%.

Autonomous Line-Follower Robot | *ATmega328P, L298N, IR Sensors* | [\[GitHub Repo\]](#) | 2025

- Assembled a differential-drive robot with hardware PWM calibration at 7.8 kHz, removing motor stalling and reducing audible noise by 90%.
- Tuned a proportional control algorithm achieving 95% accuracy on sharp 90-degree corners at competition speed.

ACHIEVEMENTS

- Secured 2nd Prize at TECHTANTRA 2K26 National Level Technical Competition among 50+ teams for the ZeroTrace project.
- Registered Indian Design Patent (Application No: 499678-001, CBR NO: 209757) in April 2026 for ZeroTrace hardware.

EDUCATION

B.Tech in Computer Science and Engineering | *S.B. Jain Institute of Technology, Nagpur* | 2023 - 2027

Specialization: IoT Architecture, Cybersecurity, Embedded Systems

Higher Secondary Certificate (HSC), Science | *Municipal Junior College, Kalmeshwar* | 2021 - 2023

Secondary School Certificate (SSC), Distinction | *Municipal Junior College, Kalmeshwar* | 2019 - 2021

CERTIFICATIONS

Cyber Security Workshop | *Crow Security / IIT Bombay* | 2025-2026

Ethical Hacking and Penetration Testing | *NASSCOM FutureSkills Prime* | 2025-2026

Security Operations Fundamentals | *Palo Alto Networks* | March 2026

Network Security Fundamentals | *Palo Alto Networks* | March 2026

Cybersecurity Fundamentals | *Palo Alto Networks* | March 2026

Cloud Security Fundamentals | *Palo Alto Networks* | March 2026

Python (Basic and Problem Solving) | *HackerRank* | 2025

AWS Cloud Computing Fundamentals | *AWS Academy* | 2025

German Language Proficiency Level 1 | *FLOA* | December 2025

LANGUAGES

English (Professional) | Hindi (Native) | Marathi (Native) | German (Level 1)